Zeno's Target: The Uncollapsable Areas

Vanishing dome–wedge geometry in the plane, on a sphere, and on a two-dimensional sphere embedded in four-dimensional space

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An exposition and synthesis; no priority claim for the underlying mathematics

Abstract

Consider a circle containing a regular polygon containing another circle, and repeat the construction indefinitely. Each annular layer splits into *domes* between the outer circle and polygon, and *wedges* between the polygon and its incircle. For a polygon with n sides, both types of area vanish like n^{-2}, but their proportions approach 2/3 and 1/3. At fixed n, infinite nesting exhausts the starting disk; letting n tend to infinity afterward produces aggregate dome and wedge areas equal to two-thirds and one-third of the original area. Reversing the limits leaves the entire disk unconsumed. We extend the construction to geodesic polygons on a round two-sphere, derive exact spherical nesting and joint-scaling laws, and show that the 2/3 limit survives curvature. We then embed the same two-dimensional construction in a higher-dimensional sphere and prove that the intrinsic result is unchanged. Finally, if the number of sides grows from layer to layer, the sharp area-exhaustion criterion is the divergence of $\sum_k n_k^{-2}$. The distinction is not between sizes of infinity but between limiting procedures and the geometric information they retain.

Keywords: circular segments, nested regular polygons, asymptotic area, geodesic polygons, spherical geometry, infinite products, noncommuting limits.

1 The target and the two vanishing regions

Zeno's dichotomy paradox concerns the completion of a journey through infinitely many subdivisions; his arrow paradox questions how motion should be understood at an instant. Here the object of subdivision is not the journey but the *target*. The question is whether geometric distinctions can survive in an aggregate even as every individual feature becomes arbitrarily small.

Start with a disk of radius R. Inscribe a regular n-gon; then inscribe the polygon's incircle. Use that incircle as the outer circle of the next layer. At each stage, define the *domes* as the portion of the outer disk outside the polygon and the *wedges* as the portion inside the polygon but outside its incircle. These are two-dimensional areas; no polyhedron is constructed. Boundaries have zero area and may be assigned to either class.

We use $n \geq 3$ for side count and $k \geq 0$ for nesting depth. Limits in these parameters will be taken separately and jointly. Our term *uncollapsable* refers to the persistence of an area *ratio or accumulated measure*; it does not mean that an individual region refuses to shrink.

2 The planar 2/3 theorem

Set $x = \pi/n$. The incircle radius is $R\cos x$. The outer disk, polygon and inner disk have areas

$$A_{\text{out}} = \pi R^2, \qquad A_{\text{pol}} = nR^2 \sin x \cos x, \qquad A_{\text{in}} = \pi R^2 \cos^2 x. \tag{1}$$

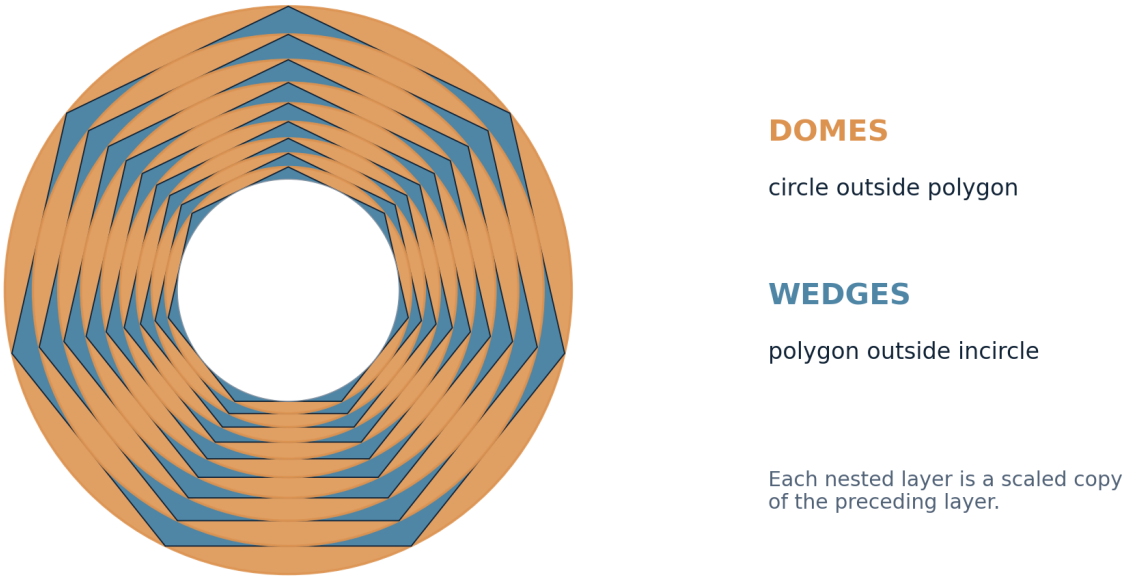


Figure 1: The planar target: nested regular polygons partition each circular annulus into orange domes and blue wedges. The innermost white disk remains after finitely many layers.

Therefore the total dome and wedge areas at one layer are

$$\mathsf{D}_n(R) = R^2\left[\pi - n\sin x\cos x\right], \tag{2}$$

$$\mathsf{W}_n(R) = R^2\left[n\sin x\cos x - \pi\cos^2 x\right], \tag{3}$$

$$\mathsf{D}_n(R) + \mathsf{W}_n(R) = \pi R^2\sin^2 x. \tag{4}$$

Theorem 1 (The uncollapsable planar proportion)**.** *For any fixed $R > 0$, the total dome and wedge areas both tend to zero as $n \to \infty$, yet*

$$\boxed{\lim_{n\to\infty}\frac{\mathsf{D}_n(R)}{\mathsf{D}_n(R)+\mathsf{W}_n(R)} = \frac{2}{3}, \qquad \lim_{n\to\infty}\frac{\mathsf{W}_n(R)}{\mathsf{D}_n(R)+\mathsf{W}_n(R)} = \frac{1}{3}.} \tag{5}$$

Proof. Taylor expansion at $x = 0$ gives

$$\mathsf{D}_n(R) = \pi R^2\left(\frac{2}{3}x^2 - \frac{2}{15}x^4 + O(x^6)\right),$$

$$\mathsf{W}_n(R) = \pi R^2\left(\frac{1}{3}x^2 - \frac{1}{5}x^4 + O(x^6)\right).$$

The leading-order coefficients are in the ratio $2:1$. The shared factor $\pi R^2 x^2$ cancels, proving the claim. □

The exact dome fraction is

$$F_n = \frac{\pi - n\sin x\cos x}{\pi\sin^2 x} = \frac{2}{3} + \frac{4\pi^2}{45n^2} + O(n^{-4}). \tag{6}$$

In other words, $\mathsf{D}_n/\mathsf{W}_n \to 2$, whereas $\mathsf{D}_n/(\mathsf{D}_n + \mathsf{W}_n) \to 2/3$. These two quantities must not be confused. The classical shallow circular-segment estimate $\mathcal{A}_{\text{segment}} \sim (2/3)(\text{chord})(\text{height})$ provides a closely related geometric explanation [1].

3 Infinite nesting and the order of infinity

Put $q_n = \cos^2(\pi/n)$, so $0 < q_n < 1$. After k completed layers, the remaining inner radius is $R\cos^k(\pi/n)$ and its area is $\pi R^2 q_n^k$. Because every layer is similar to the first, the total consumed area and its two components are exactly

$$C_{n,k} = \pi R^2(1 - q_n^k), \tag{7}$$
$$\mathsf{D}_{n,k}^{\text{tot}} = F_n \pi R^2(1 - q_n^k), \tag{8}$$
$$\mathsf{W}_{n,k}^{\text{tot}} = (1 - F_n)\pi R^2(1 - q_n^k). \tag{9}$$

Theorem 2 (Noncommuting limits)**.** *The relative area remaining obeys*

$$\boxed{\lim_{n\to\infty}\lim_{k\to\infty} q_n^k = 0, \qquad \lim_{k\to\infty}\lim_{n\to\infty} q_n^k = 1.} \tag{10}$$

Correspondingly,

$$\lim_{n\to\infty}\lim_{k\to\infty} \mathsf{D}_{n,k}^{\text{tot}} = \frac{2}{3}\pi R^2, \qquad \lim_{k\to\infty}\lim_{n\to\infty} \mathsf{D}_{n,k}^{\text{tot}} = 0. \tag{11}$$

Proof. For any finite n, $q_n^k \to 0$ as $k \to \infty$. For any finite k, $q_n^k \to 1$ as $n \to \infty$. The area statements follow from (7) and Theorem 1. □

A boundary of interpretation. The limits describe the *measures of two classes of regions*, not two fixed limiting shapes. In particular, no claim is made that individual orange and blue sets have a pointwise limit of the indicated areas. Taking the individual numerator and denominator to zero and then writing $0/0$ loses the data from which the ratio is obtained; it is not a new definition of division by zero.

4 A common clock: the joint scaling limit

For small x, $\log\cos x = -x^2/2 + O(x^4)$. Thus

$$q_n^k = \exp\left(-\frac{\pi^2 k}{n^2} + O\left(\frac{k}{n^4}\right)\right). \tag{12}$$

When $k, n \to \infty$ with $k/n^2 \to \tau \in [0, \infty)$, the remaining fraction approaches

$$\boxed{Q_{\text{flat}}(\tau) = e^{-\pi^2\tau}.} \tag{13}$$

Consequently the consumed dome and wedge areas converge to

$$\mathsf{D}_{\text{flat}}(\tau) = \frac{2}{3}\pi R^2(1 - e^{-\pi^2\tau}), \qquad \mathsf{W}_{\text{flat}}(\tau) = \frac{1}{3}\pi R^2(1 - e^{-\pi^2\tau}). \tag{14}$$

The parameter τ is not a different cardinality of infinity. It measures the relative *speed* of side refinement and nesting.

5 Moving the 2D target onto a round sphere

Now place the same two-dimensional construction on a round sphere of physical radius L. A geodesic circle of *angular radius* $a \in (0, \pi/2)$ is the boundary of a spherical cap. Edges of its inscribed regular polygon are shorter great-circle arcs. Let $x = \pi/n$. The polygon's geodesic incircle has angular radius b, determined by the right spherical triangle from the cap center to a side midpoint:

$$\boxed{\tan b = \tan a \cos x.} \tag{15}$$

The cap area of angular radius a is $2\pi L^2(1 - \cos a)$. By the spherical-excess formula (Girard's theorem), the polygon area is

$$A_{\text{pol}}^{S^2} = 2L^2 \left[\pi - n \arctan\!\big(\cos a \tan x\big)\right]. \tag{16}$$

To see (16), cut the polygon into n isosceles spherical triangles. Each base half-angle is $\beta = \pi/2 - \arctan(\cos a \tan x)$, giving a polygon area of $L^2[2n\beta - (n-2)\pi]$ [3].

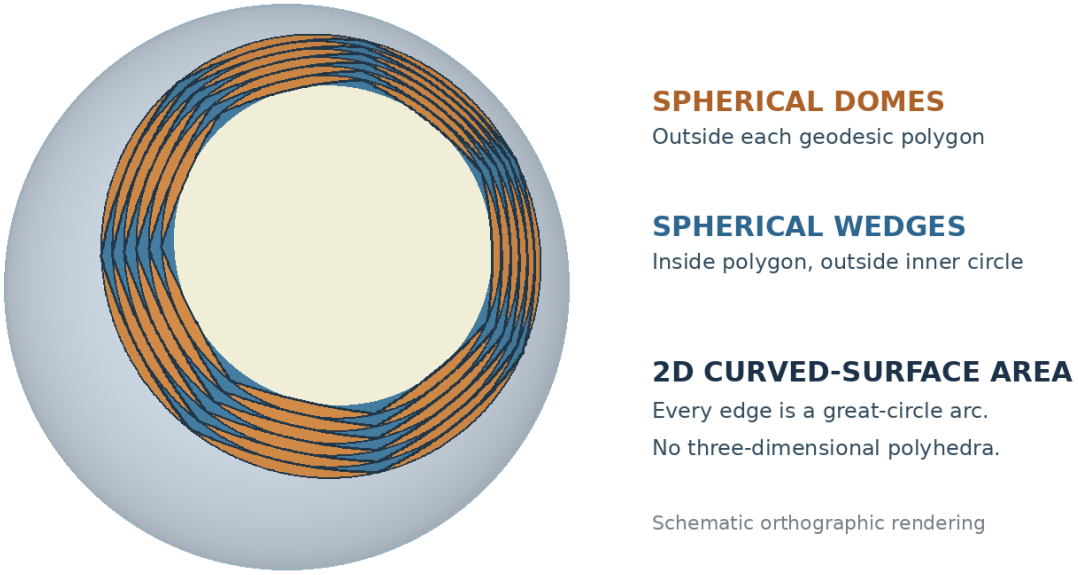


Figure 2: A genuine spherical *surface-area* construction. Great-circle polygon edges and concentric geodesic circles partition each nested annulus into domes and wedges. The projection is two-dimensional, but the measured areas are intrinsic to the curved sphere.

Define $A_{\text{out}}^{S^2} = 2\pi L^2(1 - \cos a)$ and $A_{\text{in}}^{S^2} = 2\pi L^2(1 - \cos b)$. The spherical dome and wedge areas are

$$\mathsf{D}_n^{S^2}(a) = A_{\text{out}}^{S^2} - A_{\text{pol}}^{S^2}, \qquad \mathsf{W}_n^{S^2}(a) = A_{\text{pol}}^{S^2} - A_{\text{in}}^{S^2}. \tag{17}$$

Theorem 3 (The 2/3 law survives spherical curvature)**.** *For every fixed $a \in (0, \pi/2)$ and every $L > 0$,*

$$\boxed{\lim_{n\to\infty} \frac{\mathsf{D}_n^{S^2}(a)}{\mathsf{D}_n^{S^2}(a) + \mathsf{W}_n^{S^2}(a)} = \frac{2}{3}.} \tag{18}$$

Proof. Expansion of (15) gives

$$b = a - \frac{1}{2}\sin a \cos a\, x^2 + O(x^4).$$

Equations (16)–(17) then give

$$\mathsf{D}_n^{S^2}(a) = \frac{2\pi L^2}{3}\sin^2 a \cos a\, x^2 + O(x^4),$$
$$\mathsf{W}_n^{S^2}(a) = \frac{\pi L^2}{3}\sin^2 a \cos a\, x^2 + O(x^4).$$

For $0 < a < \pi/2$ the common coefficient is positive; hence the limiting ratio is $2:1$. □

The finite-n ratio *does* depend on curvature and cap size, but the limiting one does not. At exactly $a = \pi/2$ the chosen geodesic boundary is a great circle and the construction degenerates: both areas vanish for every n and the fraction is undefined. This boundary case must not be silently included in Theorem 3.

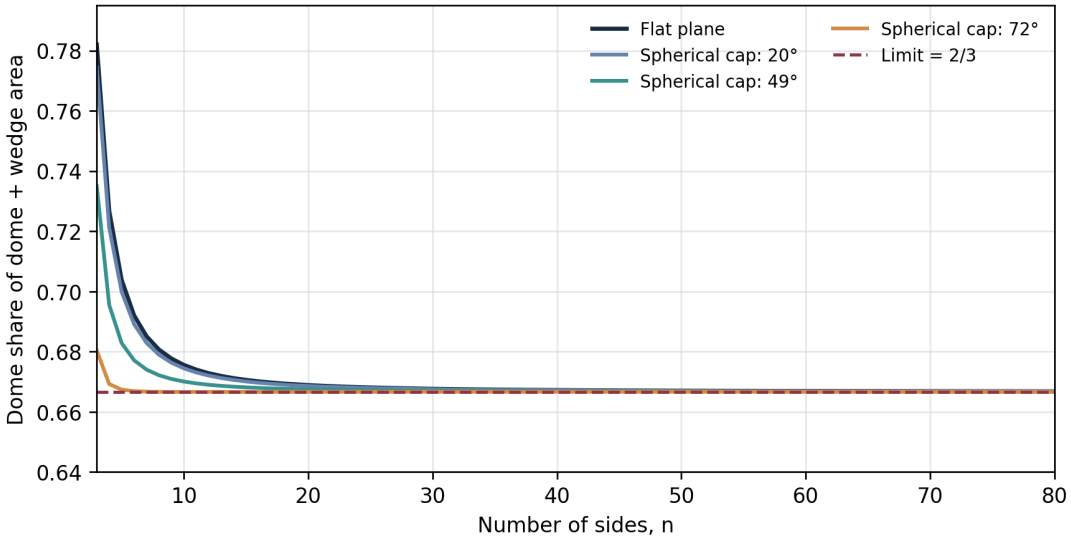


Figure 3: Numerically evaluated exact formulae: flat and spherical dome fractions approach 2/3 from different finite-side values. Spherical angular radii are below 90°.

6 Infinite spherical nesting

Let $a_0 \in (0, \pi/2)$ and keep n fixed across all layers. Repeated use of (15) yields an exact recurrence and closed form:

$$\boxed{\tan a_k = \tan a_0 \, \cos^k(\pi/n).} \tag{19}$$

The area still present after k layers is $2\pi L^2(1 - \cos a_k)$; it tends to zero for every fixed finite n. Unlike the planar case, successive layers are not exact metric copies, so one must *not* use a common geometric-series factor for dome and wedge areas individually.

Nevertheless, the single-layer fraction of Theorem 3 tends uniformly to 2/3 for $a \in (0, a_0]$ as $n \to \infty$. Weighting those fractions by each annulus's area gives

$$\boxed{\lim_{n\to\infty} \lim_{k\to\infty} \mathsf{D}_{n,k}^{S^2,\mathrm{tot}} = \frac{2}{3}\, 2\pi L^2 (1 - \cos a_0),} \tag{20}$$

while the corresponding wedge total is one-third of the starting cap area. Reversing the limits makes both totals zero, since any fixed finite number of layers consumes vanishing area as $n \to \infty$.

For joint scaling $k/n^2 \to \tau$, (19) gives

$$\tan a_\tau = \tan a_0 e^{-\pi^2 \tau/2}, \qquad a_\tau = \arctan(\tan a_0 e^{-\pi^2 \tau/2}). \tag{21}$$

The fraction of the initial spherical cap remaining is therefore

$$\boxed{Q_{S^2}(\tau) = \frac{1 - \cos a_\tau}{1 - \cos a_0}.} \tag{22}$$

Its consumed area divides into dome and wedge portions with limiting fractions 2/3 and 1/3. As $a_0 \to 0$, the spherical law reduces to the flat law $Q_{\mathrm{flat}}(\tau) = e^{-\pi^2 \tau}$.

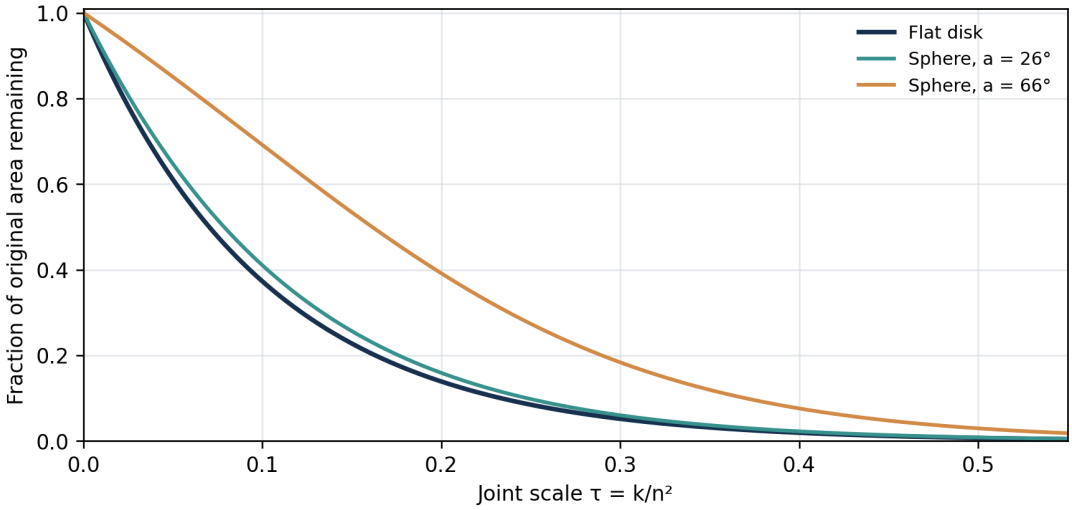


Figure 4: How much area survives when side count and nesting depth increase together. Curvature changes the decay law, although the asymptotic dome–wedge split is preserved.

7 A two-dimensional target on a four-dimensional sphere

Dimension terminology requires care. A sphere in four-dimensional Euclidean space is the three-dimensional hypersurface

$$S_L^3 = \{(x, y, z, w) \in \mathbb{R}^4 : x^2 + y^2 + z^2 + w^2 = L^2\}. \tag{23}$$

Intersect it with a three-dimensional linear subspace, for example $w = 0$. Then

$$S_L^3 \cap \{w = 0\} = \{(x, y, z, 0) : x^2 + y^2 + z^2 = L^2\} \cong S_L^2. \tag{24}$$

This is a *great two-sphere*: a two-dimensional, totally geodesic submanifold of S^3_L. Its induced metric is exactly the usual round metric on S^2_L. Consequently, the same geodesic polygons, incircles, areas, recurrence (19), and Theorem 3 apply with no alteration. No three-dimensional polygon packing, extruding, or non-touching-vertex problem arises.

Proposition 1 (Ambient-dimensional invariance)**.** *If Zeno's target is constrained to a great S^2_L inside S^m_L for any $m \geq 2$, its intrinsic dome–wedge ratios and nesting laws are the same as those on an ordinary round two-sphere. In particular, the limiting dome fraction is $2/3$.*

Proof. The intersection of $S^m_L \subset \mathbb{R}^{m+1}$ with a three-dimensional linear subspace through the origin is isometric to the round sphere S^2_L. Isometries preserve lengths, angles, geodesics and two-dimensional areas. Apply Theorem 3. □

If by "four-dimensional sphere" one means the four-dimensional *surface* $S^4 \subset \mathbb{R}^5$ instead, the same proposition applies: select a great S^2 inside S^4. Conversely, choosing an arbitrary non-round two-dimensional sheet inside a hypersphere does *not* justify reusing the exact round-sphere formulas. The intrinsic geometry, rather than the dimension of the ambient space, controls the answer.

8 Changing the number of sides at each nesting layer

An additional question supplies a sharp threshold absent from fixed-n nesting. Let layer j use $n_j \geq 3$ sides. The planar remaining area fraction after k layers is

$$Q_k = \prod_{j=1}^{k} \cos^2\left(\frac{\pi}{n_j}\right). \tag{25}$$

For a spherical target the corresponding exact formula is

$$\tan a_k = \tan a_0 \prod_{j=1}^{k} \cos\left(\frac{\pi}{n_j}\right). \tag{26}$$

Theorem 4 (Threshold for target survival)**.** *For either construction, with $n_j \geq 3$, a strictly positive area remains after infinitely many layers if and only if*

$$\boxed{\sum_{j=1}^{\infty} \frac{1}{n_j^2} < \infty.} \tag{27}$$

If this series diverges, infinitely many layers exhaust the target's area. For the spherical case we again assume $0 < a_0 < \pi/2$.

Proof. For integers $n \geq 3$ the quantity $-\log\cos(\pi/n)$ is positive and bounded above and below by positive constants times $1/n^2$; moreover, it is asymptotic to $\pi^2/(2n^2)$. Thus the logarithm of the infinite product in (25) converges to a finite value exactly when $\sum n_j^{-2}$ converges. A positive product leaves a positive radius and area in both (25) and (26). A zero product leaves none. □

Corollary 1. *If n_j grows like Cj^p for some $C > 0$, the remaining target area is positive when $p > 1/2$ and zero when $p \leq 1/2$ (with the first few n_j adjusted to be at least 3).*

Remark 1. *The $2/3$ aggregate dome fraction need not hold for an arbitrary variable-side sequence. Early coarse layers may account for a significant area and bias the total ratio. The $2/3$ result applies in the uniform large-side regime, including the fixed-n family as $n \to \infty$.*

9 What actually survives the limit?

Four conclusions should be distinguished.

1. **A local ratio survives.** The two classes of shrinking area vanish at the same quadratic order, with leading coefficients in the ratio $2 : 1$.
2. **A finite aggregate survives one ordering.** Nest completely at finite n before increasing the side count, and the total dome area tends to two-thirds of the target's original area.
3. **The operations do not commute.** Increase sides first, and a fixed number of layers consumes no area in the limit; increasing nesting afterward does not change this iterated limit.
4. **Ambient dimension is not intrinsic dimension.** A two-dimensional target on a great S^2 inside a four-dimensional ambient space has the same geometry as on an ordinary round sphere.

This is neither division by zero nor a new "higher-order infinity." It is classical analysis made visible. The $2/3$ factor is related to the familiar shallow circular-segment formula; nested polygon constructions are also well established [1, 2]. The proofs here are intended as an integrated exposition, not a novelty or priority claim.

10 Conclusion: the uncollapsable target

The destination 0 does not, by itself, encode the way a quantity approached it. The dome and wedge areas both tend to zero; the *relationship* between them does not. Under infinite nesting, that relationship can account for finite portions of the starting disk or spherical cap. On a curved surface, and even on a great two-sphere embedded in a higher-dimensional sphere, the same limiting proportion survives.

The central line of Zeno's Target is therefore best expressed precisely:

> *What vanishes as an individual region may persist as a proportion of a vanishing whole, or as the accumulated measure of infinitely many such regions.*

The target does not defeat zero. It demonstrates that *approaching* zero may retain information that the terminal value alone cannot display.

No infinities were injured in the preparation of this paper. Several were reordered without their consent.

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